UAM Corridor Communication–Transportation Simulation

Technical Model Report

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1. Overview

This report describes the mathematical and implementation-level structure of a discrete-time coupled simulation that co-models the **traffic dynamics** and **cellular communication infrastructure** of a single Urban Air Mobility (UAM) corridor. The simulator is written in Python and organized into four layers:

1. corridor/UAM kinematics
2. base-station infrastructure
3. communication networking
4. metrics calculation

Four scenarios are run to probe failure modes relevant to shared 5G infrastructure.

The simulator is intentionally minimal: it uses closed-form analytical approximations rather than physics-based flight dynamics or link-level radio models. This is appropriate for a first-pass study of *coupling* between traffic density and communication load, and for identifying the parameter regimes in which that coupling becomes safety-critical.

2. Traffic Layer

2.1 Corridor Definition

The simulation models a single one-dimensional corridor representing the San Francisco – San Jose (SF–SJ) route.

Parameter	Symbol	Value
Corridor length	L	80,000 m (80 km)
Cruise altitude AGL	h	300 m
Vertical band thickness	Δh	33 m
Number of lanes	N_ℓ	1 (baseline), 2 (high-density)
Origin hub	SF	$x = 0$ m
Destination hub	SJ	$x = L$

The corridor is modeled as a 1D segment parameterized by horizontal position $x \in [0, L]$. UAMs travel one-way from SF ($x = 0$) to SJ ($x = L$). Lateral lane separation is tracked as a discrete integer label but lane-switching logic is not implemented. When a UAM reaches $x \geq L$ it is marked as arrived and removed from all subsequent computations; no respawning is implemented in the current prototype. Future work will replace the linear segment with geographic waypoints specified in the configuration.

2.2 UAV Kinematic Model

Each UAM i carries state $(x_i, v_i, \mathrm{mode}_i, s_i^{\mathrm{BS}})$ at each time step, where x_i is longitudinal position, v_i is speed, $\mathrm{mode}_i \in \{\mathrm{HOTL}, \mathrm{HWTL}\}$ is the human–control mode, and s_i^{BS} is the index of the currently serving base station.

2.2.1 Spawn Pattern

UAMs are initialized at $t = 0$ with a **uniform spacing** pattern:

$$x_i(0) = \frac{L}{N+1} \cdot i, \quad i = 1, \ldots, N$$

where N is the number of UAMs and lane assignment is $\ell_i = i \bmod N_\ell$.

Density validation. For uniform spawn, UAMs interleave across lanes so the intra-lane gap is:

$$\Delta x_{\text{lane}} = N_\ell \cdot \frac{L}{N+1}$$

The minimum safe spacing at the slowest design speed is $v_{\min} \cdot \tau_{\min}$. The simulator raises a `ValueError` at initialization if:

$$\Delta x_{\text{lane}} < v_{\min} \cdot \tau_{\min}$$

The maximum feasible fleet size is:

$$N_{\max} = N_\ell \cdot \left\lfloor \frac{L}{v_{\min} \cdot \tau_{\min}} \right\rfloor - 1$$

Initial speed assignment. Rather than drawing speeds independently, speeds are assigned in a post-spawn pass to be consistent with the car-following model. Within each lane, UAMs are processed from leader (highest x) to last follower; each follower's initial speed is capped:

$$v_i(0) = \min\left(v_i^{\text{rand}}, \ \frac{x_{i-1}(0) - x_i(0)}{\tau_{\min}} \right)$$

where $v_i^{\text{rand}} \sim \mathcal{U}(v_{\min}, v_{\max})$. This ensures the initial state is a valid car-following configuration. In practice, for all current scenarios the uniform spacing is large enough that the cap is never active (see density validation numbers above).

Scenario	N	$v_{\min}$ (m/s)	$v_{\max}$ (m/s)	Δx_{lane} (m)	$v_{\min}\tau_{\min}$ (m)
Baseline	50	40	60	1,569	200
High density	150	35	55	1,060	105
Frequent switches	80	40	60	988	200
Cascade failure	60	40	55	1,311	200

2.2.2 Car-Following Law

The longitudinal dynamics follow a **time-gap car-following rule** applied within each lane. At each step, UAMs in a lane are sorted by position in descending order; the vehicle at the head of the queue is the *leader*.

Leader (free-flow): no vehicle ahead, constant speed:

$$x_{\text{lead}}(t + \Delta t) = x_{\text{lead}}(t) + v_{\text{lead}} \cdot \Delta t$$

Follower: let $\Delta x_i = x_{i-1}(t) - x_i(t)$ be the gap to the vehicle immediately ahead, and $\tau_{\min}$ the minimum safe time gap. The safe gap criterion is:

$$g_{\text{safe},i} = v_i \cdot \tau_{\min}$$

If $\Delta x_i < g_{\text{safe},i}$, the follower decelerates to maintain the minimum headway:

$$v_i^* = \max\left(\frac{\Delta x_i}{\tau_{\min}},\ 0\right)$$

$$v_i \leftarrow v_i^*$$

A **hard collision guard** is applied immediately after the car-following speed adjustment, before advancing position. Because the leader's position has already been updated to $x_{i-1}(t + \Delta t)$ earlier in the same loop iteration, the maximum physically

admissible speed for the follower is:

$$v_i^{\text{guard}} = \max\left(0,\ \frac{x_{i-1}(t+\Delta t) - x_i(t)}{\Delta t}\right)$$

$$v_i \leftarrow \min(v_i,\ v_i^{\text{guard}})$$

This is a strict invariant: no follower can reach or overtake its leader within a single time step, regardless of any other dynamics. Position is then advanced:

$$x_i(t+\Delta t) = x_i(t) + v_i \cdot \Delta t$$

There is no explicit maximum deceleration constraint in the current prototype; the model directly sets speed to v_i^* in one step (instantaneous response). This is a known simplification to be replaced with an IDM or similar model in future work.

Parameter	Symbol	Baseline	Cascade
Minimum safe time gap	$\tau_{\min}$	5.0 s	5.0 s
Separation violation threshold	τ_{viol}	2.0 s	2.0 s
Time step	Δt	1.0 s	1.0 s

2.2.3 Separation Metric

At each recorded step, pairwise separation times are computed for all consecutive UAM pairs within a lane:

$$T_{\text{sep},i} = \frac{x_{i+1} - x_i}{v_i}$$

where the index ordering is by ascending position ($x_{i+1} > x_i$). A **separation violation** is recorded when $T_{\text{sep},i} < \tau_{\text{viol}} = 2.0\,\text{s}$. Summary statistics include the per-step minimum and mean, plus the cumulative violation count over the run.

2.3 Human–Control Mode Switching

Each UAM operates in one of two modes at any instant:

- **HOTL** (Human-Over-The-Loop): supervisory monitoring; automation handles flight.
- **HWTL** (Human-Within-The-Loop): remote piloting via live video feed; human directly commands the aircraft.

Mode transitions follow a **Poisson process model** (default). The inter-arrival time to the next HOTL→HWTL switch for vehicle i is drawn from an exponential distribution:

$$T_{\text{switch},i} \sim \text{Exp}(\lambda), \quad \lambda = \lambda_{\text{hr}}/3600 \; [\text{s}^{-1}]$$

HWTL episode duration is independently drawn:

$$D_{\text{HWTL},i} \sim \text{Exp}(\mu_{\text{HWTL}}), \quad \mu_{\text{HWTL}} = 1/\bar{d} \; [\text{s}^{-1}]$$

Upon episode expiry, the vehicle returns to HOTL and a new inter-arrival time is drawn.

An alternative **zone-based trigger model** is also implemented: UAMs entering designated hot-zones (e.g., the first and last 5 km near hubs) switch to HWTL with per-step probability $p_{\text{zone}} \cdot \Delta t$.

Scenario	λ_{hr}	$\bar{d}$ (s)
Baseline	0.5 / UAM / hr	60
Frequent switches	5.0 / UAM / hr	45
Cascade failure	8.0 / UAM / hr	90

An optional `max_concurrent_hwtl` cap can enforce a fleet-wide HWTL concurrency limit (set to `None` in all prototype scenarios).

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3. Infrastructure Layer

3.1 Base Station Specifications

Base stations (BS) represent ground-based cellular radio access points (conceptually 5G NR gNB nodes). Each BS is modeled with a fixed capacity and a simplified **M/M/1 queuing model** for uplink congestion.

Parameter	Symbol	Baseline	High Density	Cascade Failure
Number of BS	M	16	16	6
BS capacity	C	100 Mbps	50 Mbps	20 Mbps
Good-coverage radius	r_{good}	3,000 m	3,000 m	5,000 m
Degraded-coverage radius	r_{deg}	6,000 m	6,000 m	9,000 m
Message buffer capacity	B	500 msgs	300 msgs	500 msgs

Coverage radii are measured in 3D Euclidean distance from the BS to the UAM. A UAM receives "good" service if $d \leq r_{\text{good}}$, "degraded" service if $r_{\text{good}} < d \leq r_{\text{deg}}$, and no service otherwise.

M/M/1 Queuing Model

The queuing delay at each BS is approximated with the M/M/1 mean sojourn time formula. Let λ_{arr} [bps] be the total UAM traffic arriving at the BS in a given step, and C [bps] be the BS capacity. The traffic intensity is:

$$\rho = \frac{\lambda_{\text{arr}}}{C}$$

The mean queuing delay is:

$$W = \frac{L_{\min}}{1-\rho}, \quad \rho < 1$$

where $L_{\min} = 5\,\mathrm{ms}$ is a base processing latency (serialization + scheduling overhead). For $\rho \geq 1$ (saturated), W is capped at 2.0 s to bound numerical divergence. Load factor thresholds for alerting:

Level	Threshold
Warning	$\rho \geq 0.60$
Critical	$\rho \geq 0.85$
Saturated	$\rho \geq 1.00$

Buffer overflow is checked separately: if the number of messages arriving in a step exceeds B, the excess is counted as dropped.

3.2 Infrastructure Placement

Base stations are placed at **uniform intervals along the corridor**, offset 500 m laterally from the centerline:

$$x_j^{\mathrm{BS}} = \frac{L}{M+1} \cdot j, \quad j = 1, \ldots, M$$

$$y_j^{\mathrm{BS}} = 500\,\mathrm{m}$$

This yields a spacing of:

- Baseline / high-density (16 BS): $\Delta x \approx 4{,}706\,\mathrm{m}$ ≈ 4.7 km
- Cascade failure (6 BS): $\Delta x \approx 11{,}429\,\mathrm{m}$ ≈ 11.4 km

The 3D Euclidean distance from BS j to UAM i at position x_i and altitude $h = 300\,\mathrm{m}$ is:

$$d_{ij} = \sqrt{(x_j^{\mathrm{BS}} - x_i)^2 + (y_j^{\mathrm{BS}})^2 + h^2}$$

Note that UAM lateral position (within-lane) is projected onto the corridor centerline ($y_{\mathrm{UAM}} = 0$) for coverage computation; only longitudinal position is resolved. Each BS is also permanently connected to the Ground Control Station (GCS) via a lossless backhaul link with fixed RTT = 10 ms.

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4. Network Layer

4.1 Communication Type

The communication substrate is modeled as a **cellular uplink/downlink** network, abstracting a 5G NR-like air–ground link. No explicit radio propagation model (path loss, fading, SINR) is implemented; connectivity is a deterministic threshold on 3D distance. This is a first-order approximation consistent with the prototype scope.

Communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$:

- $\mathcal{V}$: UAM nodes, BS nodes, one GCS node
- $\mathcal{E}$: directed edges (u, BS) and (BS, u) for each connected UAM; edges $(\mathrm{BS}, \mathrm{GCS})$ always present

Each edge carries attributes: `quality` ∈ {good, degraded}, `rtt` (seconds), `rtt_ok` (bool).

4.2 Messages Exchanged

Each UAM generates two classes of messages per time step, independent of mode:

Basic Safety Message (BSM)

Conceptually aligned with ASTM F3548 / GUTMA UTM message formats:

Field	Value
Payload size	300 bytes
Broadcast rate	10 Hz
Bit rate	300 × 8 × 10 = **24,000 bps** (24 kbps)
Purpose	Position, velocity, identity; shared with all network nodes

Control / Telemetry Stream

The uplink/downlink control channel has two bandwidth regimes depending on mode:

Mode	Payload	Bit Rate	Purpose
HOTL	Supervisory telemetry	**35 kbps**	Aircraft health, automation status; human monitors only
HWTL	Live video + command	**6 Mbps**	Full video feed to ground pilot; human actively commands

Total per-UAM demand on serving BS:

$$R_i = R_{\text{BSM}} + R_{\text{ctrl}} = \begin{cases} 59\,\text{kbps} & \text{mode} = \text{HOTL} \\ 6{,}024\,\text{kbps} \approx 6\,\text{Mbps} & \text{mode} = \text{HWTL} \end{cases}$$

This ~100× bandwidth ratio between modes is the primary driver of network stress: a single HWTL aircraft consumes as much bandwidth as roughly 100 HOTL aircraft.

Per-step message count: At $\Delta t = 1\,\text{s}$, each UAM generates 10 BSM messages and 10 control messages = **20 messages/step/UAM** at the serving BS. This count is used for buffer-overflow tracking independent of bit-rate accounting.

External (Terrestrial) Load

In the Cascade Failure scenario, non-UAM ground users (modeled as a stadium event) inject additional traffic on all BS simultaneously:

$$R_{\text{ext}}(t) = \begin{cases} 8\,\text{Mbps} & 120\,\text{s} \leq t \leq 360\,\text{s} \\ 0 & \text{otherwise} \end{cases}$$

This load competes for BS bandwidth with UAM traffic but is not attributed to the UAM message queue (terrestrial users have logically separate queues; only the bandwidth competition effect is modeled).

4.3 RTT Computation

End-to-end round-trip time for UAM i served by BS j is composed of three additive terms:

$$\text{RTT}_i = 2\left(t_{\text{prop},ij} + W_j + t_{\text{backhaul}}\right) + \delta_{\text{handoff}}$$

Component	Formula / Value	Notes
Propagation delay	$t_{\text{prop}} = d_{ij}/c$, $c = 3 \times 10^8$ m/s	Dominated by altitude; $d \approx 300$–3000 m → $t_{\text{prop}} \approx 1$–10 µs
Queuing delay	$W_j = L_{\min}/(1 - \rho_j)$	M/M/1; grows without bound as $\rho \to 1$
Backhaul delay	$t_{\text{backhaul}} = 10\,\text{ms}$	Fixed BS–GCS link latency
Handoff penalty	$\delta_{\text{handoff}} = 100\,\text{ms}$ (if in handoff)	Added once during active handoff period

The RTT safety threshold is $\text{RTT}_{\max} = 500\,\text{ms}$. Violations ($\text{RTT} > \text{RTT}_{\max}$) are flagged per UAM per step; the maximum fleet-wide RTT is tracked as a KPI.

4.4 Handoff Between Base Stations

Serving BS assignment and handoff follow a nearest-BS-with-hysteresis rule executed at every time step.

Step 1 — Coverage scan. For each active UAM, compute d_{ij} to every BS and identify the candidate set:

$$\mathcal{C}_i = \{j : d_{ij} \leq r_{\text{deg}}\}$$

If $\mathcal{C}_i = \emptyset$, the UAM is in a coverage gap; all graph edges to that UAM are removed.

Step 2 — Best BS selection. Sort $\mathcal{C}_i$ by distance; the nearest BS is the candidate j^*.

Step 3 — Hysteresis check. If UAM i is already associated with BS j_{prev} and $j^* \neq j_{\text{prev}}$, a handoff is triggered only if the candidate is at least 20% closer than the current server:

$$d_{ij^*} \leq 0.8 \cdot d_{i,j_{\text{prev}}}$$

If this condition is not met and j_{prev} still provides coverage, the UAM stays on j_{prev}. This prevents ping-pong at cell edges.

Step 4 — Handoff execution. When a handoff is triggered:

- The old BS–UAM edges are removed from $\mathcal{G}$.
- `in_handoff = True` for a period $T_{\text{HO}} = 200\,\text{ms}$.
- During T_{HO}: each message is independently dropped with probability $p_{\text{drop}} = 0.1$; RTT is penalized by $+100\,\text{ms}$.
- After T_{HO} elapses, the UAM associates cleanly with j^* and new edges are added.

Step 5 — Graph update. New directed edges (u_i, j^*) and (j^*, u_i) are written with `quality` set to "good" or "degraded" based on d_{ij^*} vs. r_{good}.

Parameter	Symbol	Value
Handoff execution delay	T_{HO}	200 ms

Parameter	Symbol	Value
Message drop probability during handoff	p_{drop}	0.10
RTT penalty during handoff	δ_{HO}	100 ms
Hysteresis margin	—	20% distance reduction required

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5. Simulation Execution

The simulator advances in uniform discrete time steps $\Delta t = 1\,\text{s}$ over a horizon $T = 600\,\text{s}$ (10 minutes). Each step executes five phases in strict order:

1. **Position update** — car-following law applied to all UAMs.
2. **Mode switching** — Poisson timers decremented; HOTL↔HWTL transitions executed.
3. **Coverage and handoff** — serving BS re-evaluated; handoff timers decremented.
4. **Traffic and queuing** — per-BS traffic loads computed; M/M/1 delay and buffer overflow evaluated; RTT computed per UAM.
5. **KPI recording** — all time-series metrics written to collector.

All scenarios use a seeded pseudo-random number generator (`random.Random(seed)`) for reproducibility.

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6. Simulation Scenarios

#	Name	N	M	C [Mbps]	λ_{hr}	External Load	Classification
1	Baseline	50	16	100	0.5	None	Normal
2	High Density	150	16	50	0.5	None	Unsafe
3	Frequent Switches	80	16	100	5.0	None	Normal
4	Cascade Failure	60	6	20	8.0	+8 Mbps @ t=120–360 s	Unsafe

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7. Known Limitations and Future Work

The following are deliberate simplifications in the current prototype, flagged for replacement with validated sub-models:

Component	Current Model	Planned Replacement
UAV kinematics	Instantaneous speed adjustment, no dynamics	IDM or Wiedemann car-following; eVTOL thrust/drag model
Safe separation time	$\tau_{\min}$ = 2–5 s (placeholder)	Calibrated from Joby S4 / Toyota NX performance envelope
Radio propagation	Binary threshold on 3D distance	A2G path-loss model (free-space + terrain shielding)
Link capacity	Fixed BS capacity	SINR-based spectral efficiency; frequency reuse
5G queuing	M/M/1	M/M/1/K or M/D/1; multi-class priority queuing for UAM vs. ground users
Corridor geometry	1D linear	SF–SJ geographic waypoints, no-fly zones, multi-lane 3D
Network topology	Static BS placement	Dynamic HAPS or moving-relay topology
HWTL switch trigger	Poisson (stationary)	Event-triggered: weather, airspace conflict, ATC instruction

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Appendix A: Key Equations Summary

Equation	Description
$d_{ij} = \sqrt{(x_j^{\mathrm{BS}} - x_i)^2 + (y_j^{\mathrm{BS}})^2 + h^2}$	3D UAM–BS distance
$\rho_j = \sum_{i \in S_j} R_i \ / \ C_j$	BS traffic intensity (utilization)
$W_j = L_{\min}/(1-\rho_j),\ \rho_j < 1$	M/M/1 mean queuing delay
$\mathrm{RTT}_i = 2(d_{ij}/c + W_j + t_{\mathrm{bh}}) + \delta_{\mathrm{HO}}$	Round-trip time
$T_{\mathrm{sep},i} = (x_{i+1} - x_i)/v_i$	Longitudinal separation time
$v_i^* = \max(\Delta x_i/\tau_{\min},\ 0)$	Car-following speed adjustment
$T_{\mathrm{switch}} \sim \mathrm{Exp}(\lambda_{\mathrm{hr}}/3600)$	Poisson mode-switch inter-arrival
$D_{\mathrm{HWTL}} \sim \mathrm{Exp}(1/\bar{d})$	HWTL episode duration

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Appendix B: Default Parameter Table

Parameter	Symbol	Default Value
Simulation horizon	T	600 s
Time step	Δt	1.0 s
Corridor length	L	80,000 m
Corridor altitude	h	300 m AGL
UAM speed range	$[v_{\min}, v_{\max}]$	[40, 60] m/s
Min separation time	$\tau_{\min}$	5.0 s
Violation threshold	τ_{viol}	2.0 s

Parameter	Symbol	Default Value
BS spacing (baseline)	Δx	≈ 4,706 m
BS capacity (baseline)	C	100 Mbps
Good-coverage radius	r_{good}	3,000 m
Degraded-coverage radius	r_{deg}	6,000 m
Buffer capacity	B	500 msgs
BSM size	—	300 bytes
BSM rate	—	10 Hz
HOTL control rate	R_{HOTL}	35 kbps
HWTL video rate	R_{HWTL}	6 Mbps
RTT safety threshold	RTT_{max}	500 ms
Base processing delay	$L_{\min}$	5 ms
Backhaul delay	t_{bh}	10 ms
Handoff duration	T_{HO}	200 ms
Drop probability (handoff)	p_{drop}	0.10
Default switch rate	λ_{hr}	0.5 / UAM / hr
Default HWTL duration	$\bar{d}$	60 s
Load warning threshold	—	60%
Load critical threshold	—	85%